

PROJECT DESIGN PHASE

Solution Architecture

Date	01 JULY 2026
Team ID	
Project Name	Strategic Product Placement Analysis
Maximum Marks	2 Marks

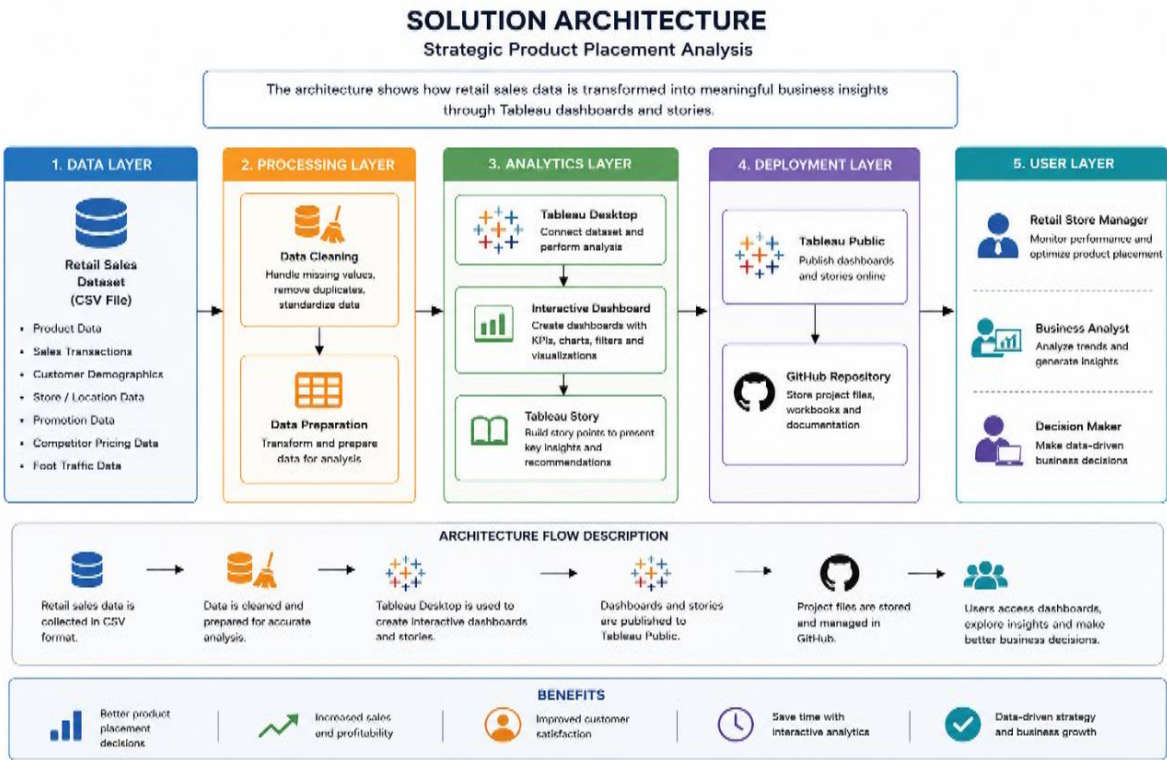
Introduction

The **Solution Architecture** defines the overall structure and workflow of the **Strategic Product Placement Analysis** project. It illustrates how retail sales data is collected, processed, analyzed, visualized, and presented to support business decision-making.

The architecture begins with the retail sales dataset stored in a CSV file. The dataset is imported into Tableau Desktop, where it undergoes data preparation and visualization. Interactive dashboards, KPI cards, filters, and Tableau Stories are created to generate business insights. Finally, the dashboard is published on Tableau Public, while all project files and documentation are maintained using GitHub.

The proposed architecture ensures an efficient, scalable, and user-friendly business intelligence solution for retail analytics.

Solution Architecture Workflow



Architecture Components

Component	Description
Retail Sales Dataset	Contains product positioning, pricing, promotions, customer demographics, foot traffic, seasonal information, and sales volume.
Data Cleaning & Validation	Removes inconsistencies, validates records, and prepares clean data for analysis.
Data Preparation	Organizes fields, creates calculated measures, and formats data for Tableau.
Tableau Desktop	Performs data analysis and develops interactive dashboards.
Interactive Charts & KPIs	Displays sales trends, category analysis, product position analysis, and business metrics.
Dashboard Development	Combines multiple visualizations into an interactive analytical dashboard.
Tableau Story	Presents business insights in a logical sequence using story points.
Tableau Public	Publishes dashboards online for easy sharing and accessibility.
GitHub Repository	Stores project reports, documentation, screenshots, and Tableau workbook.
Retail Store Manager / Business Analyst	Uses dashboard insights to improve product placement and retail decision-making.

Data Flow within the Architecture

Step	Process	Output
1	Collect Retail Sales Dataset	Product_Positioning.csv
2	Clean and Validate Data	Processed Dataset
3	Prepare Data for Analysis	Structured Data
4	Import into Tableau Desktop	Data Source
5	Create Charts & KPIs	Visualizations
6	Build Interactive Dashboard	Dashboard
7	Create Tableau Story	Story Points

Step	Process	Output
8	Publish on Tableau Public	Online Dashboard
9	Upload Project to GitHub	Documentation & Source Files

Users and Responsibilities

User	Responsibilities
Retail Store Manager	Reviews dashboards and improves product placement strategies.
Business Analyst	Analyzes sales performance and prepares business insights.
Marketing Team	Studies promotional effectiveness and customer trends.
Inventory Manager	Uses sales analysis for inventory planning.
Management	Makes strategic business decisions based on dashboard insights.

Advantages of the Solution Architecture

Advantage	Description
Interactive Analytics	Easy visualization of retail sales data.
Data-Driven Decisions	Supports informed business strategies.
Scalable Architecture	Can handle additional retail datasets.
User-Friendly Interface	Easy navigation using Tableau dashboards.
Centralized Reporting	All insights available in a single dashboard.
Online Accessibility	Dashboard accessible through Tableau Public.
Documentation Support	GitHub stores project files and reports securely.

Expected Outcome

The proposed solution architecture provides a systematic approach for transforming raw retail sales data into meaningful business insights. By integrating Tableau Desktop, Tableau Public, and GitHub, the system enables efficient data analysis, interactive visualization, and proper project documentation. The architecture helps retailers optimize product placement, understand customer purchasing behaviour, improve sales performance, and make data-driven business decisions.

Conclusion

The **Strategic Product Placement Analysis Solution Architecture** provides a structured framework for processing retail sales data and generating actionable business insights. Through the use of Tableau Desktop, interactive dashboards, Tableau Stories, and online publishing, the system simplifies retail analytics and enhances decision-making. The architecture is scalable, efficient, and suitable for supporting future enhancements in retail business intelligence.